

GNSS Spoofing Detection Performance Report

Final Model Rankings & Performance:

XGBoost (Top Performer): Achieved an outstanding accuracy of **95.38%** with a recall of **96.22%** for spoofing signals. It demonstrated superior stability and the best AUC-ROC score of **0.9899**.

Random Forest: Delivered strong results with **93.25%** accuracy, providing clear insights into feature importance (e.g., **Cnr_L1** and **Pr_Error**).

Deep Neural Network (DNN): Stable performance with approximately **90%** accuracy after 100 epochs of training.

DNN_1_Model

Epoch 100: val_auc did not improve from 0.95504

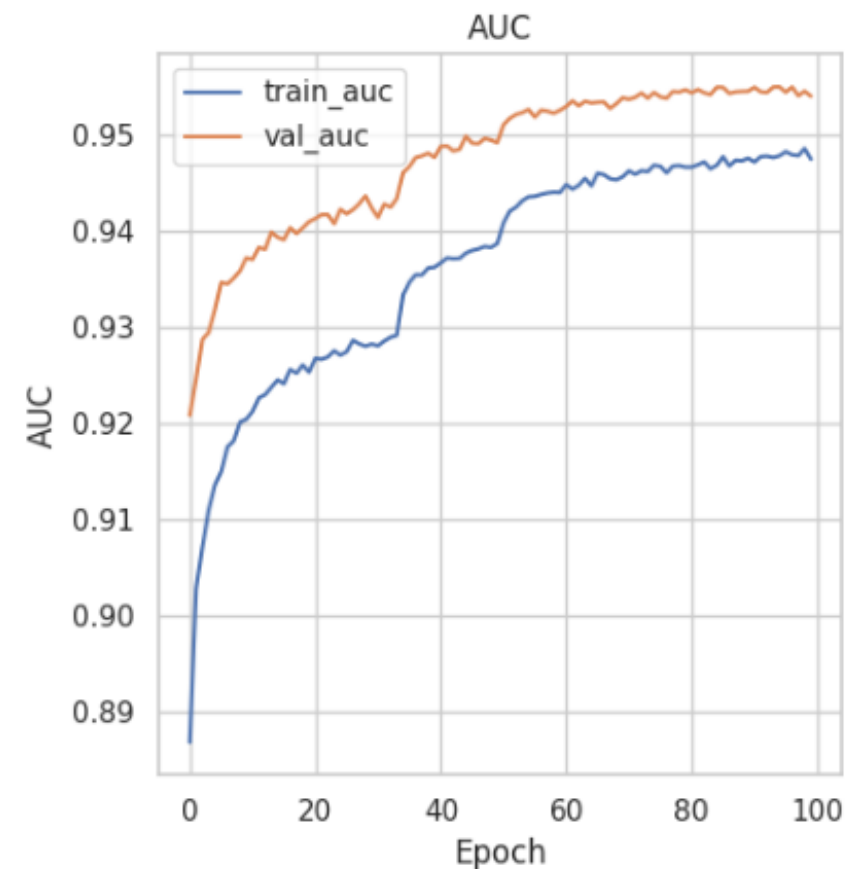
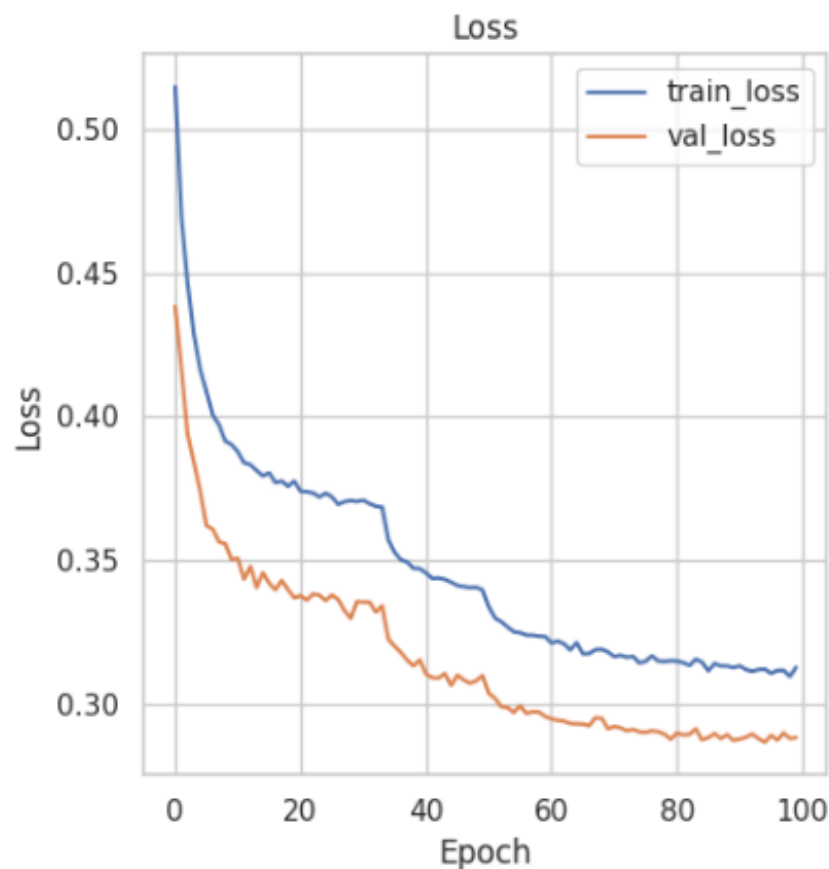
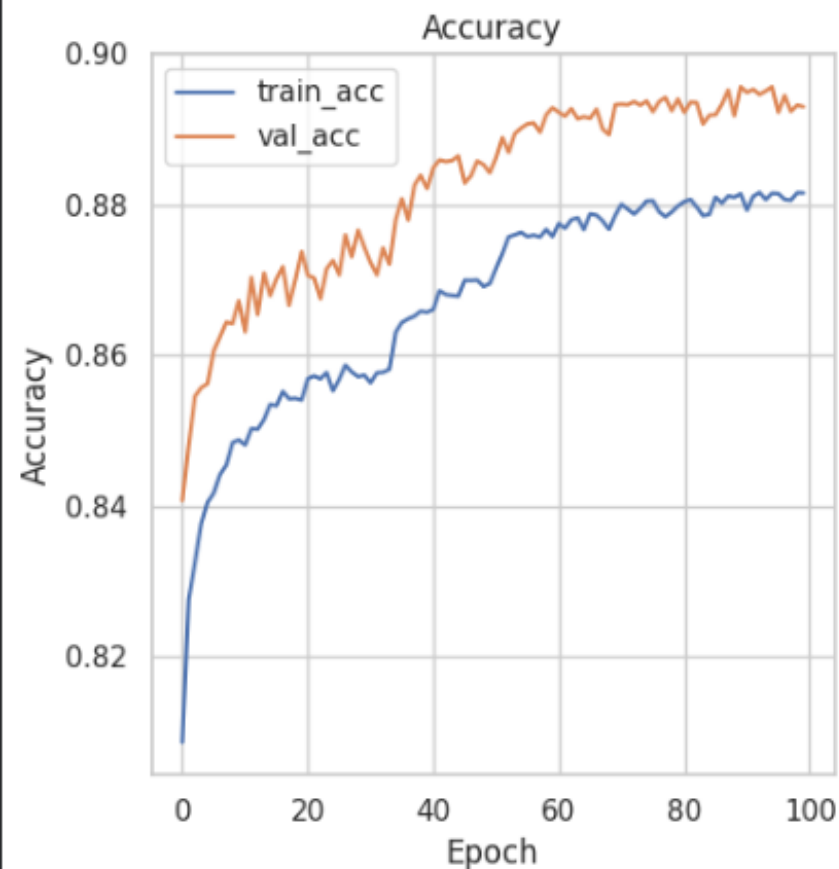
1853/1853 — 11s 6ms/step - accuracy: 0.8800 - auc: 0.9461 - loss: 0.3163 - precision: 0.9133 - recall: 0.8830 - val_accuracy: 0.8930 - val_auc: 0.9541 - val_loss: 0.2880

Restoring model weights from the end of the best epoch: 95.

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save\_model(model)`. This file format is considered legacy. We recommend using instead the SavedModel format via `tf.saved\_model.save(model, save\_dir)`.

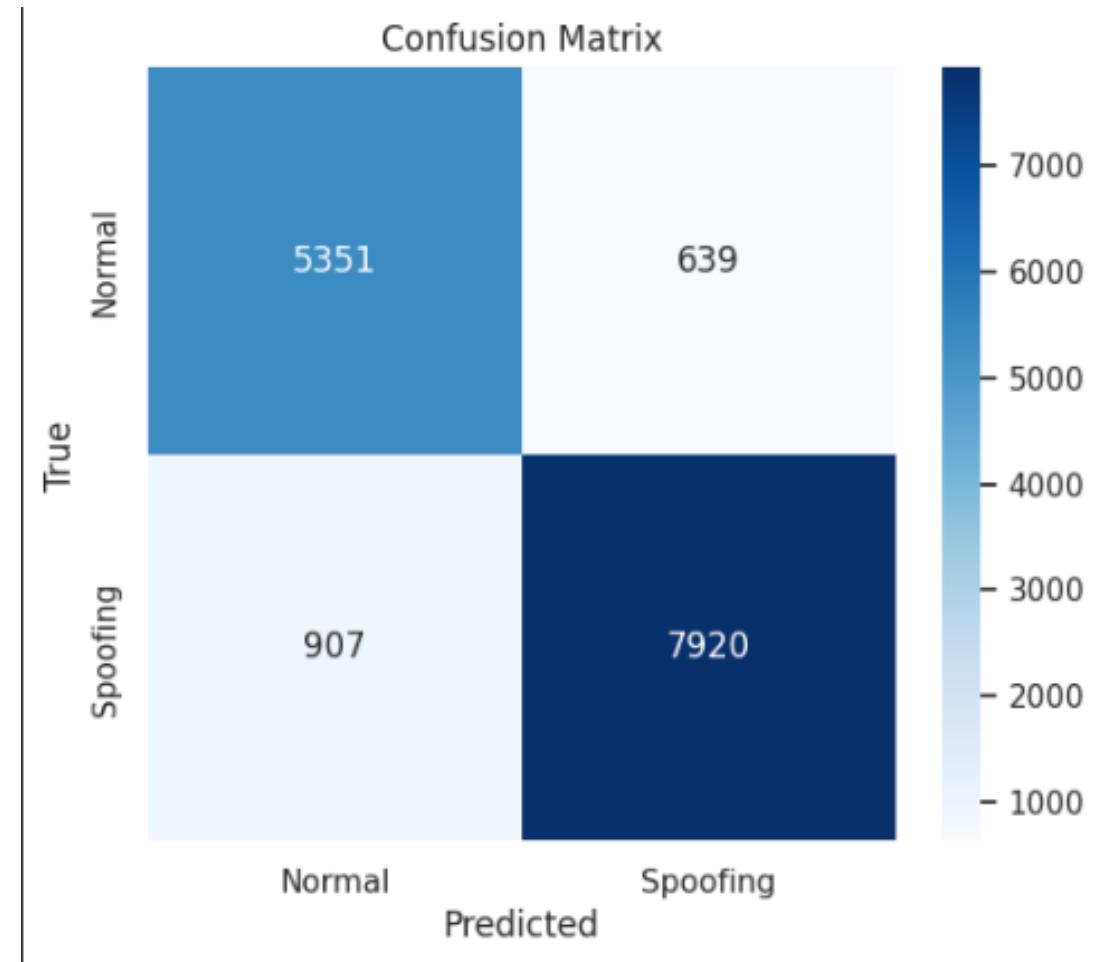
Final model saved as gnss_deep_model.h5

Best model (by val_auc) saved as gnss_deep_best.h5

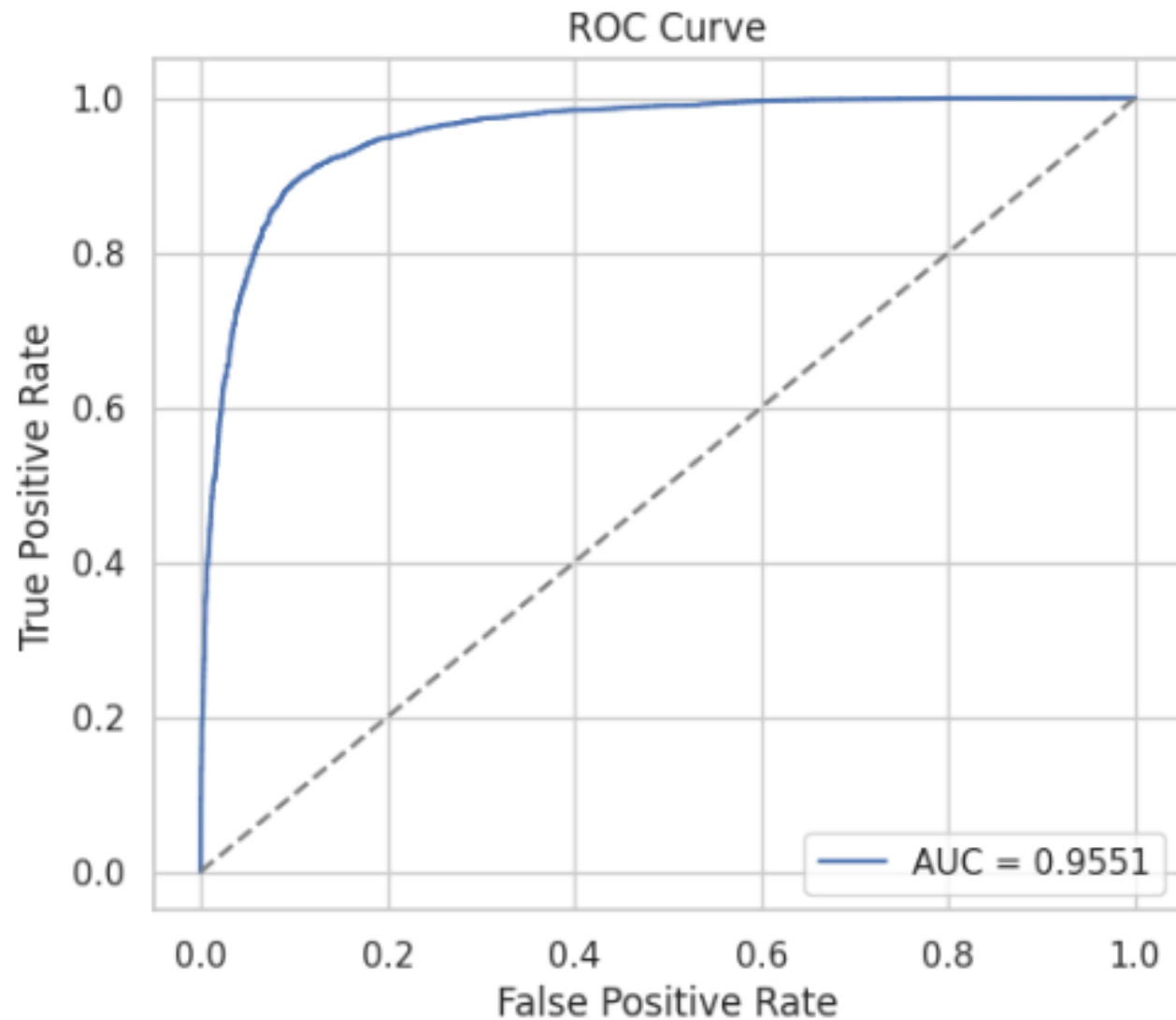


Confusion Matrix

Classification Report:				
	precision	recall	f1-score	support
0	0.8551	0.8933	0.8738	5990
1	0.9253	0.8972	0.9111	8827
accuracy			0.8957	14817
macro avg	0.8902	0.8953	0.8924	14817
weighted avg	0.8969	0.8957	0.8960	14817



ROC AUC: 0.9551

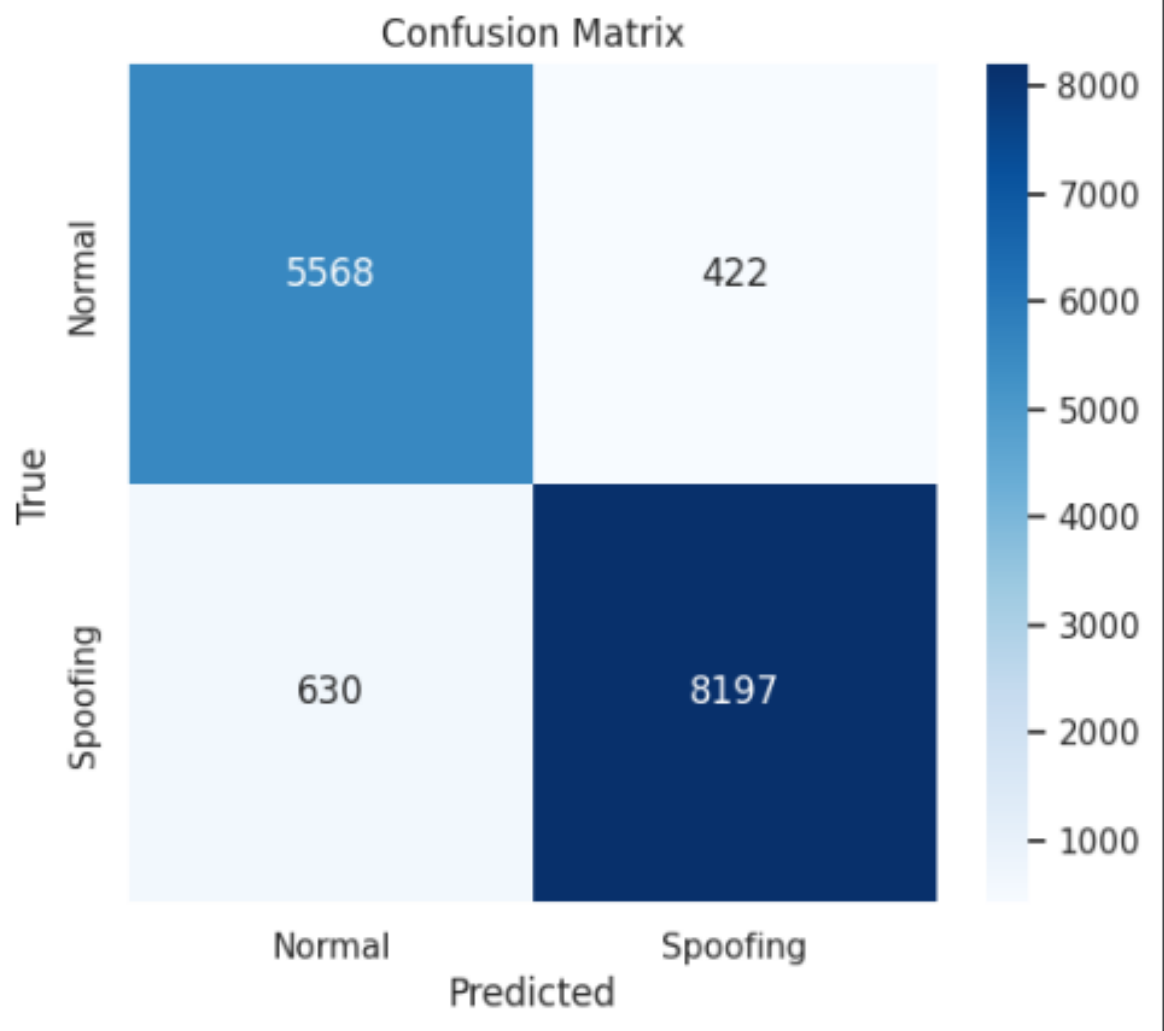


Scaler (again) saved as scaler_dnn.pkl
Training history saved to training_history.csv

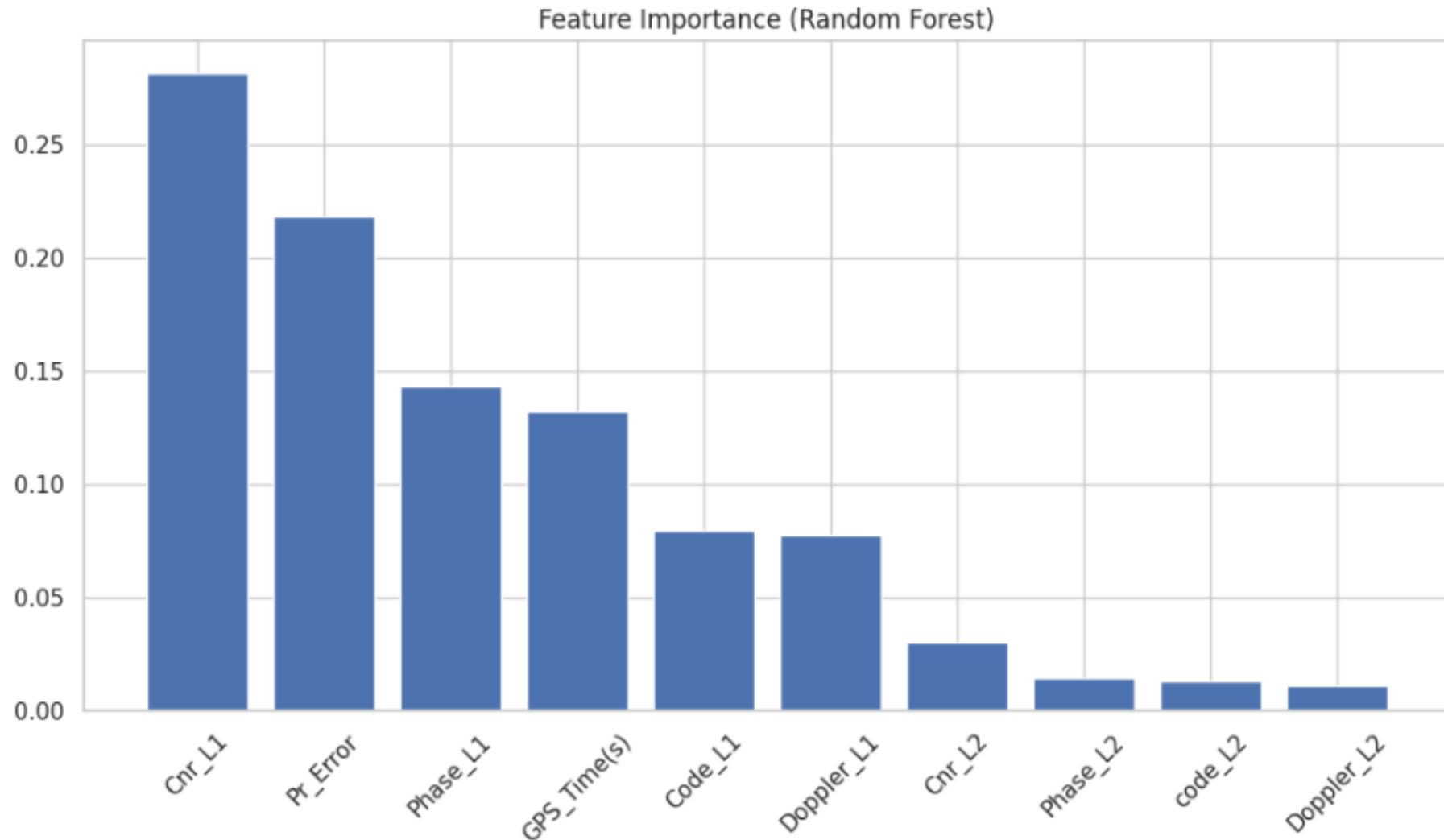
ROC Curve



RF_2_Model

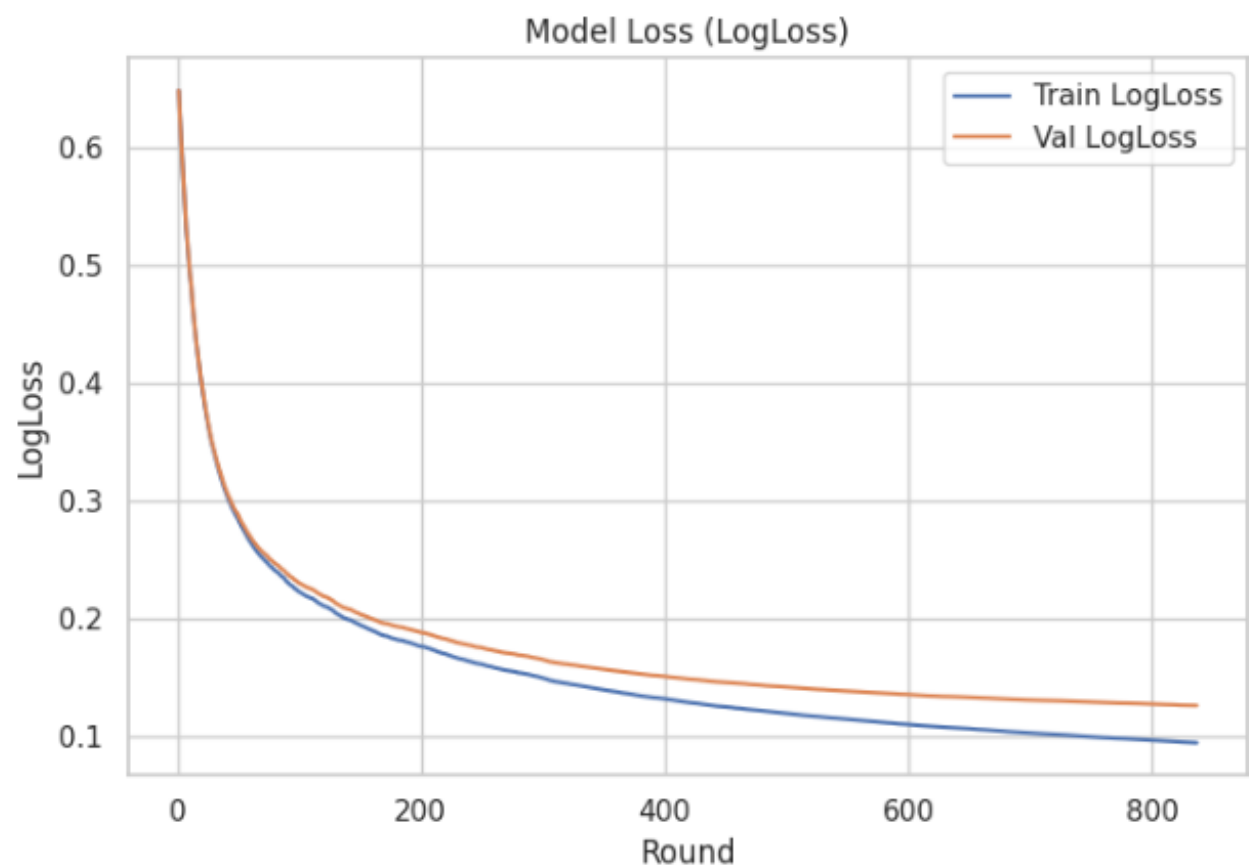
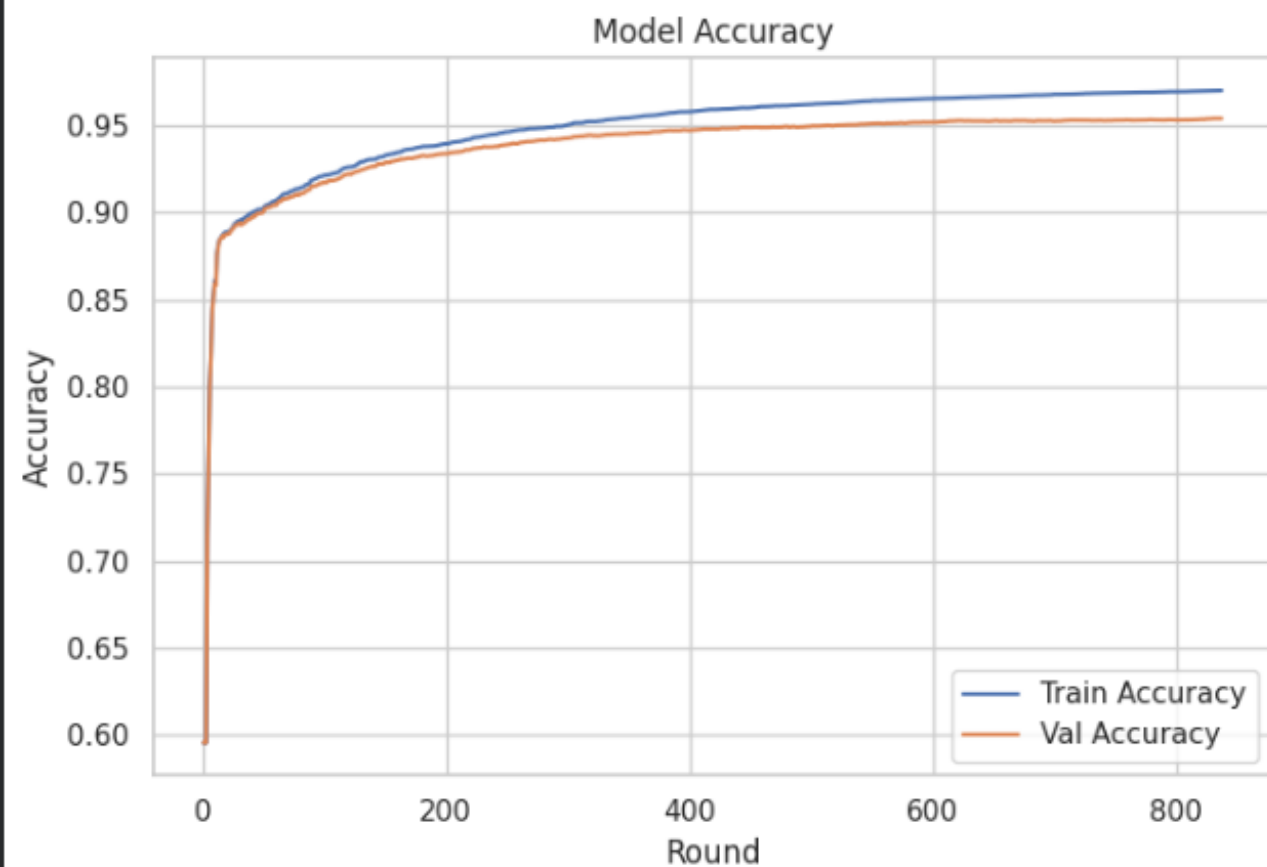


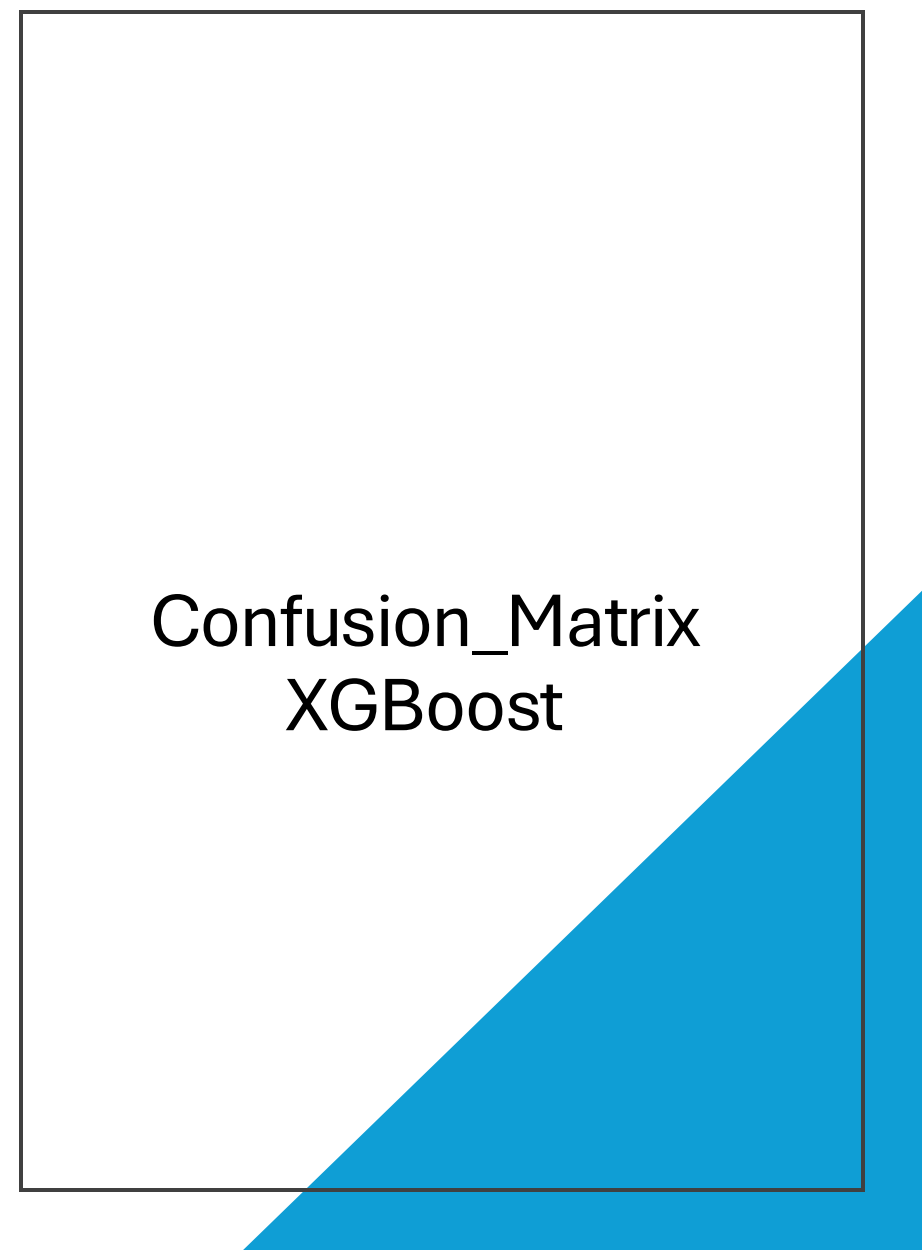
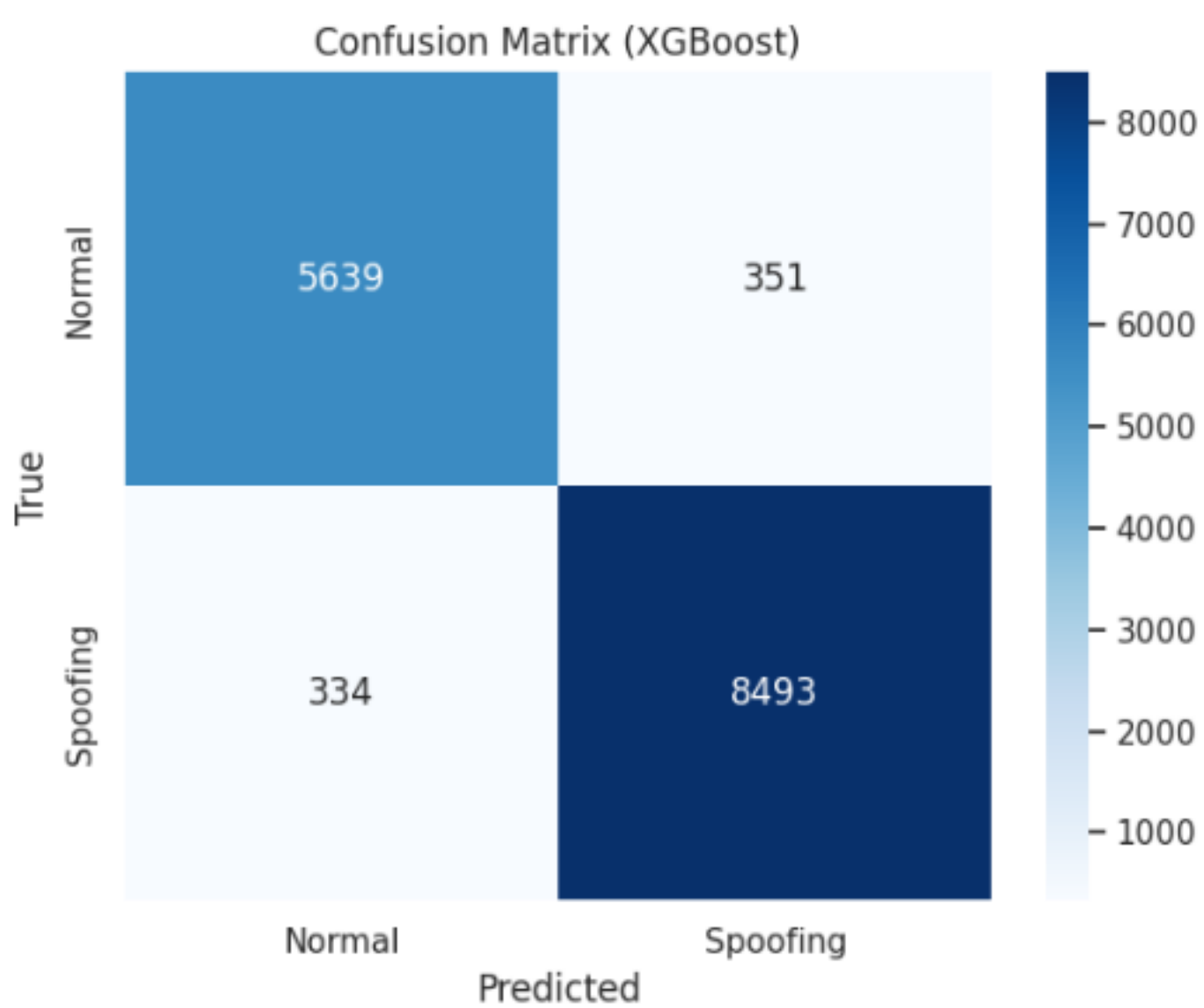
Feature importance(RF)



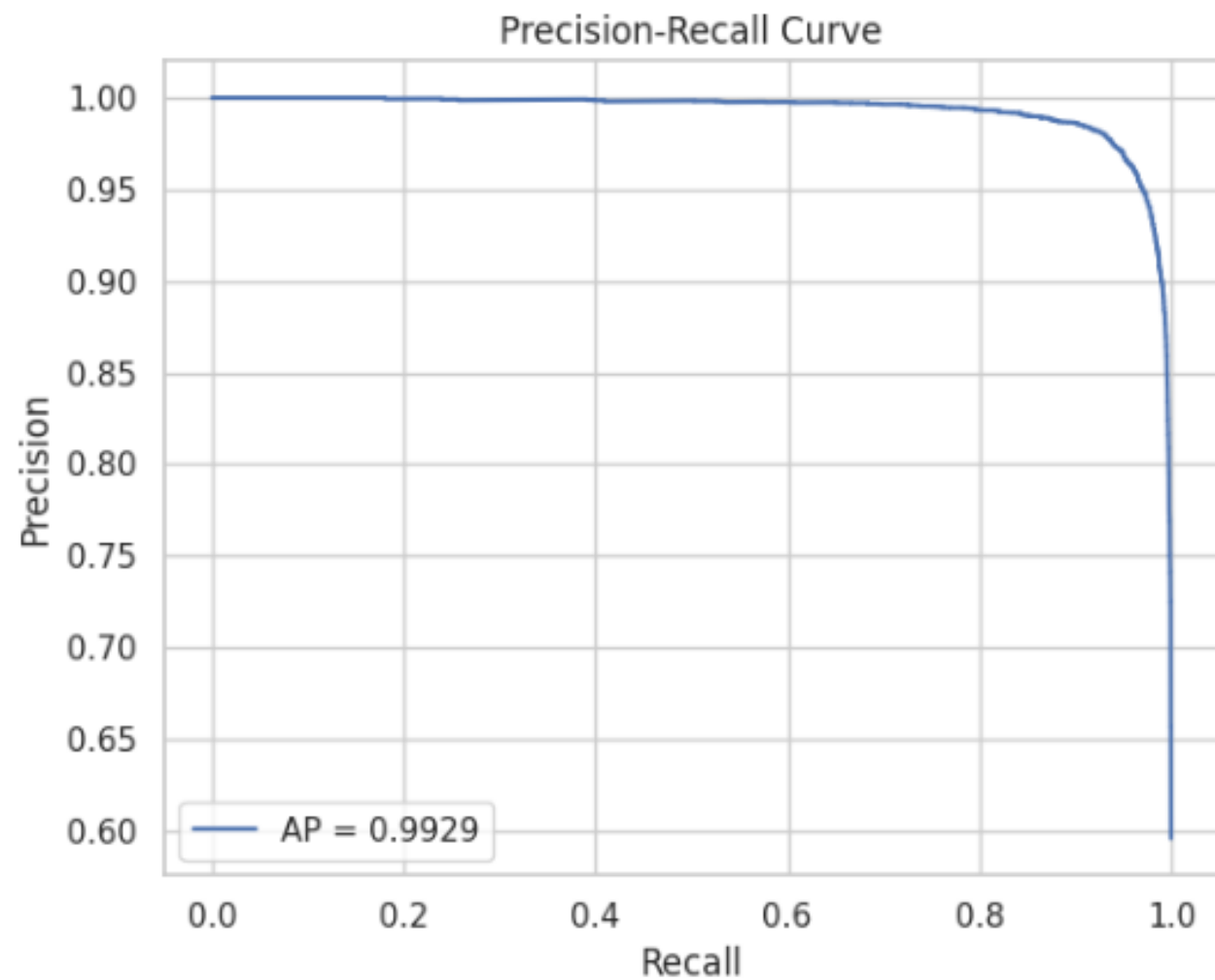
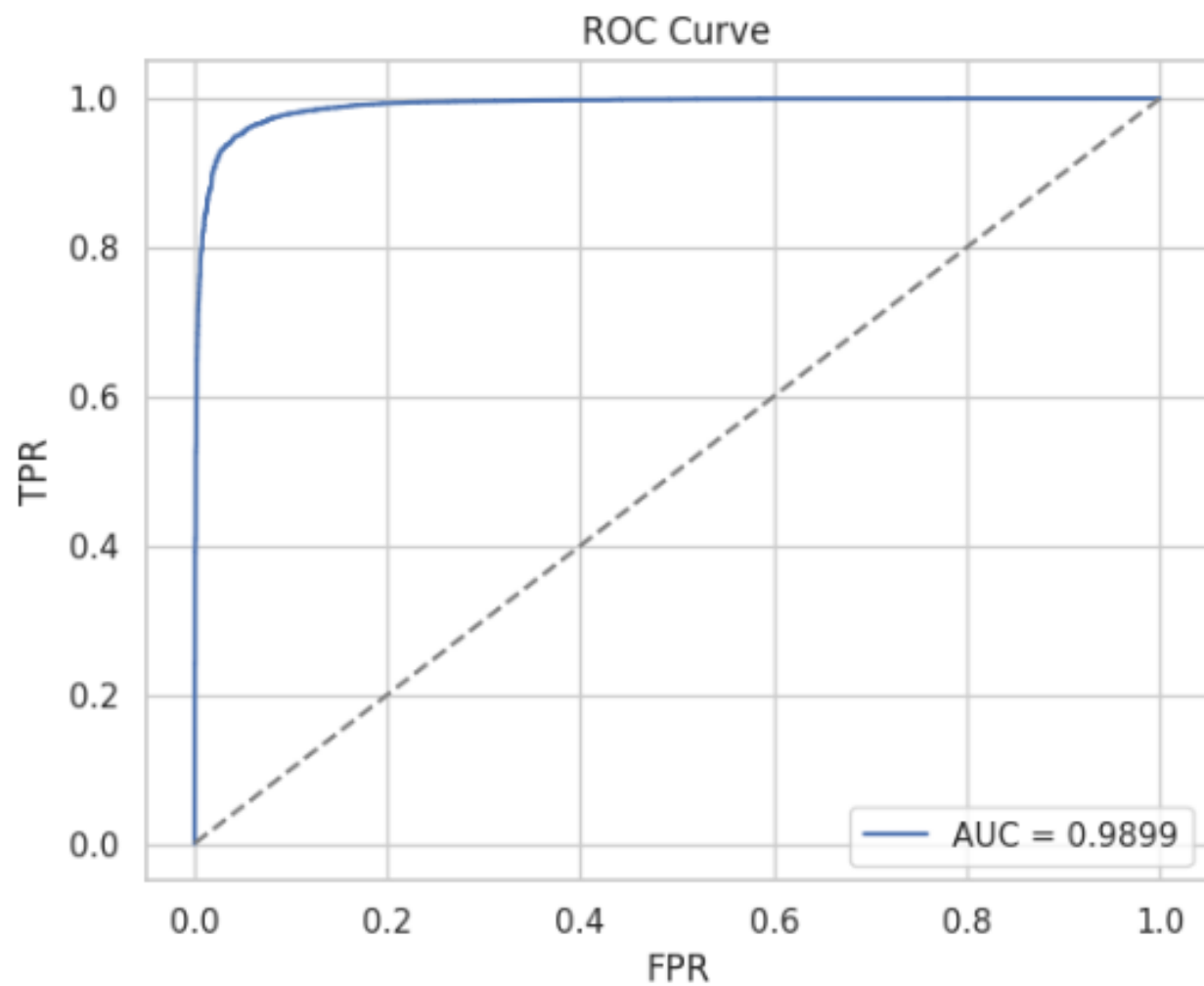
XGBoost_3_Model

```
Epoch 833/837 - accuracy: 0.9697 - loss: 0.0940 - val_accuracy: 0.9538 - val_loss: 0.1255  
Epoch 834/837 - accuracy: 0.9697 - loss: 0.0939 - val_accuracy: 0.9538 - val_loss: 0.1255  
Epoch 835/837 - accuracy: 0.9698 - loss: 0.0939 - val_accuracy: 0.9538 - val_loss: 0.1255  
Epoch 836/837 - accuracy: 0.9698 - loss: 0.0939 - val_accuracy: 0.9538 - val_loss: 0.1255  
Epoch 837/837 - accuracy: 0.9698 - loss: 0.0938 - val_accuracy: 0.9538 - val_loss: 0.1254
```

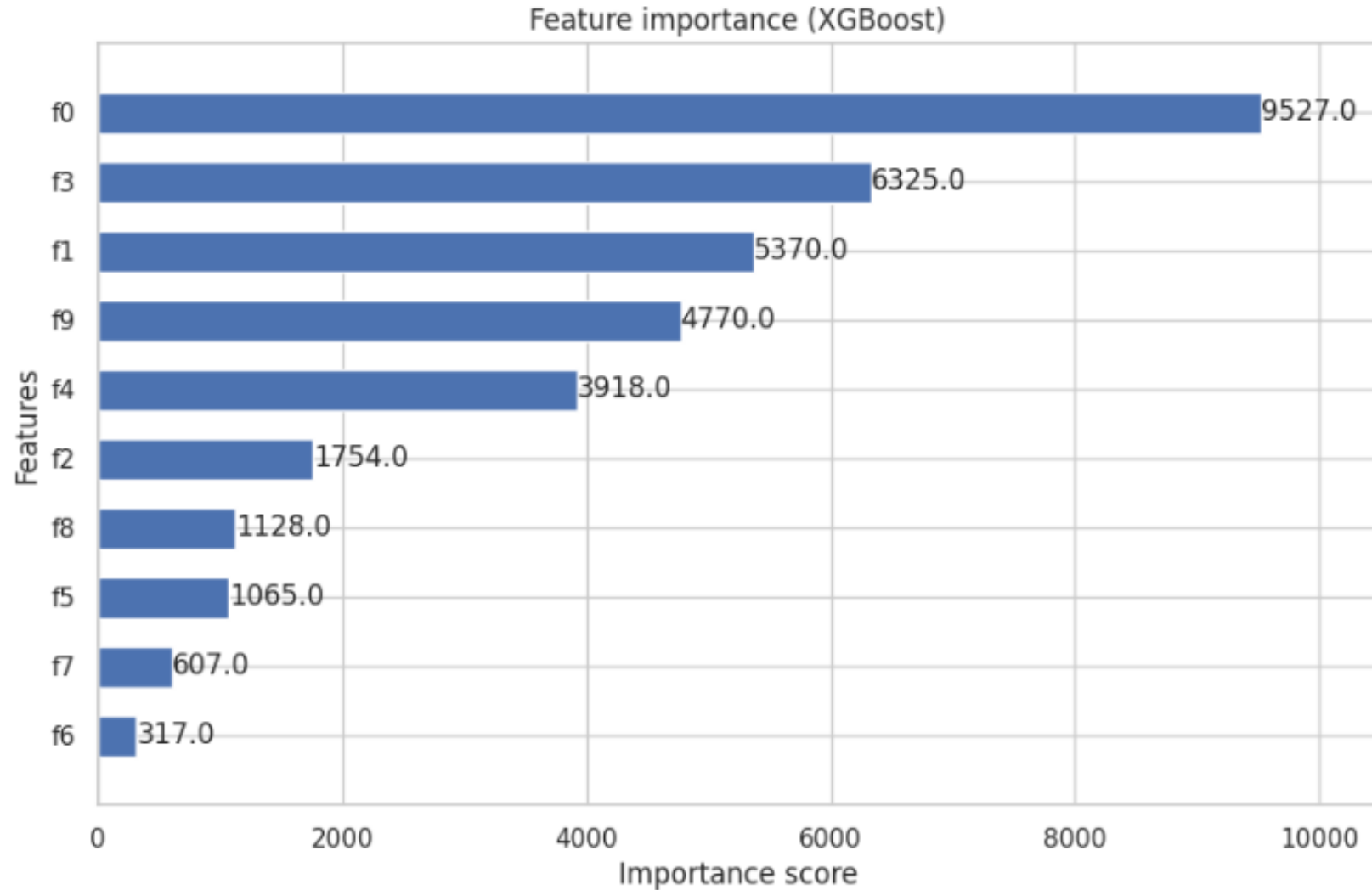




ROC Curve and Precision



Feature Importance(XGBoost)



THANK YOU